

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 2, Study for the Treatment of Anemia With Alpha (?) -Thalassemia to Determine the Efficacy and Safety of Luspatercept (BMS-986346/ACE-536) in Adults and Evaluate the Safety and Pharmacokinetics in Adolescents
Short Title:	A Study to Determine the Efficacy and Safety of Luspatercept in Adult Participants and to Evaluate the Safety and Pharmacokinetics in and Adolescent Participants With Alpha (?) -Thalassemia
Protocol Number:	154123056028960064
NCT Number:	NCT05664737
Sponsor Name:	Bristol Myers Squibb
Investigational Product:	luspatercept
Phase:	PHASE2
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Number of participants with ? 50% reduction from baseline in RBC transfusion burden with a reduction of at least 2 units during any continuous 12 weeks during Week 13-48 compared to 12-week interval immediately prior to date of first dose: Adult TD Cohort 2. Number of participants with an increase from baseline of ? 1.0 grams (g)/decilitre (dL) in mean hemoglobin (Hb) values over the continuous 12-week interval from Week 13 to Week 24 in the absence of RBC transfusion: Adult NTD Cohort 3. Dose-limiting toxicities (DLTs) defined as observance of ? Grade 3-related hemolytic crises or ? Grade 3-related event outside of the known safety profile occurring within the 21 days from their first dose of study therapy: Adolescent TD and NTD Cohorts
Secondary Obj.:	1. Number of participants with ? 33% reduction from baseline in RBC transfusion burden with a reduction of at least 2 units during any continuous 24-week interval on treatment compared to 24-week interval immediately prior to date of first dose 2. The longest duration with ? 50% reduction from baseline in RBC transfusion burden with a reduction of at least 2 units 3. Number of RBC transfusion units from week 1 to week 48

2.2 Background & Rationale

The purpose of the study is to evaluate the efficacy and safety of luspatercept plus best supportive care (BSC) vs placebo plus BSC on anemia in adult participants with ?-thalassemia hemoglobin H (HbH) disease and determine the safety and drug levels in adolescent participants.

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE2, Status: RECRUITING
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Target N: 249

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

* Adult participant? 18 years with documented diagnosis of A-Thal HbH disease with Transfusion dependence defined as:.

1. TD participant: ? 6 RBC units during the 24 weeks prior to randomization.

2. NTD participant: \< 6 RBC units during the 24 weeks prior to randomization(transfusion due to conditions other than A-Thal will not be considered)and, RBC transfusion-free during at least 8 weeks prior to randomization(unless transfusion was required to treat an acute medical condition other than A-Thal) and, mean baseline Hb ? 10 g/dL, based on a minimum of 2 measurements ? 1 week apart within 4 weeks prior to randomization; hemoglobin values within 21 days post-transfusion will be excluded.

* Adult participant has Eastern Cooperative Oncology Group (ECOG) 34 score of 0 or 1.

* Adolescent participant 12 years to \< 18 years with documented diagnosis of A-Thal HbH disease with transfusion dependence defined as:.

1. TD participant: ? 4 RBC events during the 24 weeks prior to enrollment and, no transfusion-free period for \> 56 days during the 24 weeks prior to enrollment. Participants must have a history of regular transfusions for at least 2 years.

2. NTD participant: \< 4 RBC events during the 24 weeks prior to enrollment and RBC transfusion-free during at least 8 weeks prior to enrollment and, mean baseline Hb ? 10 g/dL, based on a minimum of 2 measurements ? 1 week apart within 4 weeks prior to enrollment, hemoglobin values within 21 days post-transfusion will be excluded.

3. Participant has Karnofsky (age ?16 years) or Lansky (age \< 16 years) performance status score ? 50 at screening.

Key Exclusion Criteria:

* Medical Conditions: Diagnosis of A-ThalTrait, Hb Bart hydrops, ATRx A-Thal, hemoglobin S/?-thalassemia, myelodysplasia subtype anemia, or with HbE homozygous beta gene mutation. Anemia related to nutritional deficiency, anemia of chronic disease, autoimmune hemolytic anemia, or any other hemolytic anemias. Undergone episodes of hemolysis not related to A-Thal within the 8 weeks prior to randomization.

* Participant has deep vein thrombosis (DVT), stroke or other thromboembolic event(s) (except clogged indwelling catheter) requiring medical intervention ? 24weeks prior to randomization.

* Participant has uncontrolled hypertension. Controlled hypertension for this protocol is considered: blood pressure value corresponding to ?Grade 1 according to NCI CTCAE Version 5.0. with or without pharmacological treatment.

* Reproductive Status: Women who are pregnant, plan to get pregnant during the study, or who are breastfeeding.

* Prior/Concomitant: Undergone HSCTs or gene therapy (candidates for HSCT or gene therapy with waiting period of ? 12 months are eligible).

* Use of hydroxyurea treatment ? 12 weeks prior to enrollment for NTD participants and ? 24 weeks for TD participants.

* Participant who has extramedullary hematopoiesis (EMH) complications requiring treatment to control the growth of EMH mass(es) during the screening period.

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* Any medical or psychiatric condition (including active infections, recent surgery, sequelae of diseases or interventions, clinically significant laboratory abnormalities or concurrent treatment) that in the opinion of the investigator would put the participant at unacceptable risk of participating in the study or that could affect interpretability of data.

* Other protocol-defined inclusion/exclusion criteria apply.

4.2 Exclusion Criteria

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11. Study Identifiers & Governance

Primary ID: 154123056028960064

Registry Number: NCT05664737

Sponsor: Bristol Myers Squibb

Study Title: A Study to Determine the Efficacy and Safety of Luspatercept in Adult Participants and to Evaluate the Safety and Pharmacokinetics in and Adolescent Participants With Alpha (?) -Thalassemia

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____